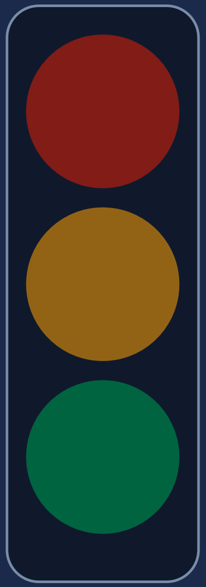
Traffic Light

Decision Controller



EEE120 Digital Design Fundamentals — Final Project

Group 4 | Usmon Shamirzayev & Team

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Smart Cities · Traffic Management · Embedded Systems

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Problem Statement

The Challenge

Small road crossings lack intelligent signal control. Without logic-based systems:

- Emergency vehicles get delayed
- Pedestrians have no safe crossing phase
- Side roads are ignored when main is empty
- Fixed timers waste time when roads are clear

Our Solution

Design a combinational logic circuit that automatically decides which signal turns green based on real-time road conditions — no clock, no memory, pure logic.

Real-World Relevance

- Smart Cities
- Autonomous Vehicles
- IoT Traffic
- Urban Infrastructure
- Embedded Systems
- Road Safety

System Overview

INPUTS

- **M** Main Road Traffic
- **S** Side Road Traffic
- **P** Pedestrian Button
- **E** Emergency Vehicle

LOGIC CONTROLLER

Boolean Logic
(AND · OR · NOT)

OUTPUTS

- **Main Green** — Main road = GREEN
- **Side Green** — Side road = GREEN
- **Ped Walk** — Pedestrians ALLOWED

Logic Design — Boolean Expressions

1 Emergency (E=1)

Overrides ALL.
Main road cleared.
Everything else RED.

2 Pedestrian (P=1)

Ped Walk ON.
Both roads RED.

3 Side Only (S=1,M=0)

Side road GREEN.
Main stays RED.

4 Default

Main road GREEN.
(default state)

Boolean Expressions

Main_Green = E + (E'· P'· (M + S')) Side_Green = E'· M'· P'· S Ped_Walk = E'· P

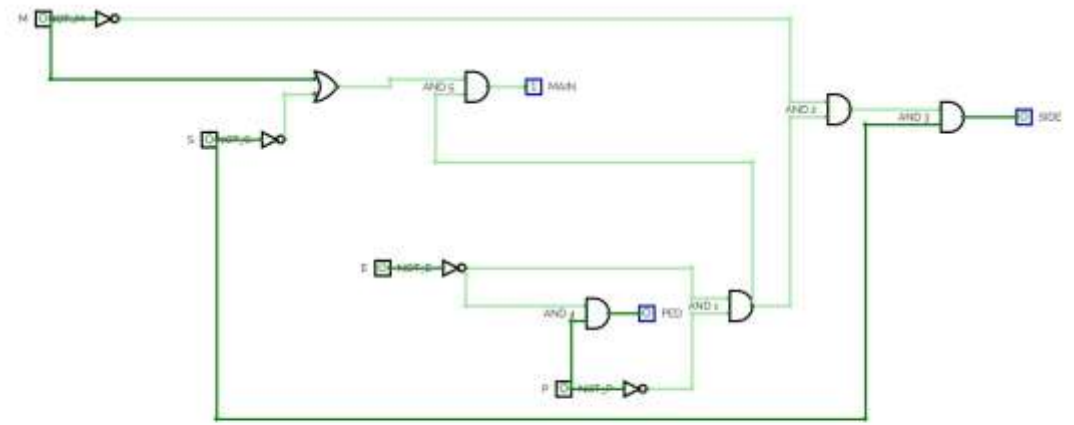
' = NOT · = AND + = OR | Gates: 3× NOT + 4× AND + 4× OR = 11 total ✓

Truth Table

M	S	P	E	Main	Side	Ped	State / Reason
0	0	0	0	1	0	0	Default → Main Green
0	1	0	0	0	1	0	Side only → Side Green
0	0	1	0	0	0	1	Ped pressed → Walk
1	0	0	0	1	0	0	Main traffic → Main Green
1	1	0	0	1	0	0	Both traffic → Main priority
1	1	1	0	0	0	1	Ped overrides traffic
0	0	0	1	0	0	0	Emergency → All RED
1	1	1	1	0	0	0	Emergency overrides all

Showing 8 key rows of 16 total combinations. Full table available in README.

CircuitVerse Design



Circuit Components

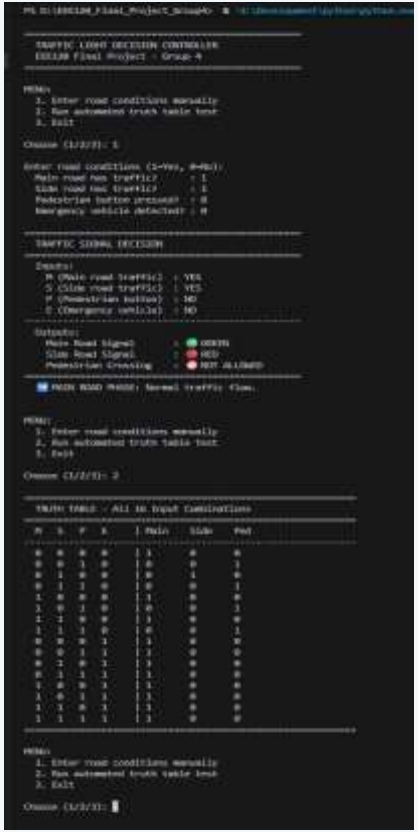
- **INPUT (×4)**
  M, S, P, E
- **NOT (×3)**
  E', M', P'
- **AND (×4)**
  Combine signals
- **OR (×4)**
  Final outputs
- **OUTPUT (×3)**

Total: 11 gates ✓
(min. requirement: 5)

circuitverse.org/users/426843/projects/eee120_final_group4

Python Software Prototype

```python
# Boolean logic — matches circuit exactly
def traffic_logic(M, S, P, E):
  NOT_E = 1 - E
  NOT_M = 1 - M
  NOT_P = 1 - P
  NOT_S = 1 - S
  main_green = E | (NOT_E & NOT_P
                   & (M | NOT_S))
  side_green = NOT_E & NOT_M & NOT_P & S
  ped_walk   = NOT_E & P
  return main_green, side_green, ped_walk
```



Manual input mode
4 road conditions

Auto truth table
All 16 combos

Live signal display
Emoji output

Explains reasoning
For each decision

Python 3
No external libs

AI / LLM Usage Reflection

□ We used Claude (Anthropic) as an AI assistant — AI was NOT used to fully make the project

Task	What AI helped with	What we understood & verified
Output formatting	Improve readability and clarity of program output	We adjusted output to clearly display signal states and meanings
Code comments & explanation	Generated structured comments explaining each code part	We reviewed all comments and verified they correctly describe the logic
Debugging & problem solving	Assisted with Python syntax and logic errors	We tested each fix and verified correctness through multiple test cases
Presentation preparation	Helped structure and format the presentation slides	We reviewed, edited, and ensured the presentation reflects our understanding

Conclusion & Future Work

✓ What Worked

- Boolean logic correctly controls all 3 outputs
- Python matches circuit logic exactly
- Emergency override works in all test cases
- Clean CircuitVerse simulation with labels

⚠️ Challenges

- Setting correct gate priorities
- Handling conflicting inputs (P=1 and E=1)
- Mapping Python operators to logic gates
- Keeping circuit readable in CircuitVerse

Future Improvements

- Sequential logic with flip-flops for timed phases
- 4-way intersection support with more inputs
- Arduino/Raspberry Pi physical implementation
- LCD countdown timer display
- Night mode — flashing yellow

Thank You! — EEE120 Group 4 — Traffic Light Decision Controller